DGCA AVIATION METEOROLOGY

METAR & TAF WEATHER CODE MASTER DECODER — FOR DAIVAM MEHTA

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1. Anatomy of a METAR String

Example: METAR VIDP 241200Z 27012G22KT 240V300 4000 R28/1400D +TSRA FEW015CB BKN030 28/24 Q1008 NOSIG=

Code Element	Decoded Value	Operational Pilot Meaning
VIDP	New Delhi IGI Airport	ICAO 4-letter station indicator
241200Z	Day 24, 1200 UTC (Z)	Observation timestamp in UTC (IST = UTC + 5:30)
27012G22KT	Wind 270° True at 12 kt, Gust 22 kt	Runway crosswind calculations required
240V300	Varying 240° to 300°	Wind direction varying by >= 60° (speed >= 3 kt)
4000	Prevailing Visibility 4000 m	General horizontal optical visibility
R28/1400D	RVR Rwy 28 is 1400m, Decreasing	Runway Visual Range (U=up, D=down, N=no change)
+TSRA	Heavy Thunderstorm with Rain	Hazardous flight conditions in vicinity
FEW015CB	1-2 oktas CB at 1,500 ft AGL	Cumulonimbus present (Severe turbulence/icing)
BKN030	5-7 oktas base at 3,000 ft AGL	Ceiling established at 3,000 ft AGL
28/24	Temp: 28°C / Dewpoint: 24°C	Spread is 4°C (Cloud base ~ 1600 ft AGL)
Q1008	QNH: 1008 hPa	Altimeter setting (sub-scale rounded down)
NOSIG	No Significant Change expected	TREND valid for 2 hours following report

2. Master Weather Descriptors & Phenomenon Matrix

Intensity / Proximity	Descriptor	Precipitation	Obscuration / Other
- Light Moderate (no sign) + Heavy VC Vicinity (8-16 km)	MI Shallow BC Patches PR Partial DR Low Drifting BL Blowing SH Showers TS Thunderstorm FZ Freezing	DZ Drizzle RA Rain SN Snow SG Snow Grains PL Ice Pellets GR Hail (>= 5mm) GS Small Hail / Graupel UP Unidentified	FG Fog (vis < 1000m) BR Mist (vis 1000-5000m) HZ Haze (RH < 80%) FU Smoke VA Volcanic Ash DU Dust SA Sand SQ Squall | SS Sandstorm

3. The 4 Strict Criteria for CAVOK

CAVOK replaces visibility, RVR, weather phenomena, and cloud groups ONLY when ALL FOUR are met:

1. Visibility is **10 km or more** (9999).
2. **NO CLOUD** below 5,000 ft (1,500 m) or below highest Minimum Sector Altitude (MSA), whichever is greater.
3. **NO Cumulonimbus (CB)** and **NO Towering Cumulus (TCU)** at any level.
4. **NO significant weather phenomena** at or around the aerodrome.

4. 10 DGCA Real-World Exam Drills Decoded

Drill 1: VABB 150600Z 09004KT 060V120 2000 HZ NSC 31/19 Q1012 NOSIG=

Mumbai: Wind 090°/04 kt varying 060°-120°. Vis 2000 m in Haze (HZ, RH < 80%). No Significant Cloud (NSC). Temp 31°C, Dewpoint 19°C. QNH 1012 hPa. No significant change expected.

Drill 2: VECC 020900Z 18015KT 3000 -RA SCT018 BKN030 26/24 Q1004 BECMG 5000 NSW=

Kolkata: Wind 180°/15 kt. Vis 3000 m in light rain. Clouds: 3-4 oktas at 1800 ft, 5-7 oktas ceiling at 3000 ft. Becoming visibility 5000 m, No Significant Weather (NSW).

Drill 3: VOBL 101200Z 00000KT 6000 FEW025CB SCT030 27/22 Q1016 TEMPO 2000 +TSRA=

Bengaluru: Calm wind. Vis 6000 m. 1-2 oktas CB at 2500 ft, 3-4 oktas at 3000 ft. Temp 27°C, DP 22°C. QNH 1016 hPa. Temporarily: Vis dropping to 2000 m in heavy thunderstorm with rain.

Drill 4: VOMM 050000Z 32008KT 0800 R07/1000U FG VV002 21/21 Q1014 BECMG 2000 BR=

Chennai: Vis 800 m in Fog (FG). RVR Rwy 07 is 1000 m and upward/increasing (U). Sky obscured, vertical visibility 200 ft (VV002). Becoming vis 2000 m in mist (BR).

Drill 5: VIDP 180300Z 00000KT 0150 R28/0350V0600D FZFG BKN001 M01/M01 Q1022=

Delhi Winter: Calm wind. Vis 150 m. RVR Rwy 28 varying 350 m to 600 m, decreasing (D). Freezing Fog (FZFG). Ceiling 100 ft. Sub-zero Temp: -1°C, Dewpoint: -1°C (M = minus).